

Exhibit 5.28 provides further detail. When using all five asset classes (the black efficient frontier), the optimal asset allocation supports a 3.61 percent withdrawal rate over thirty years with a 10 percent chance for failure. The optimal asset allocation is shown. Meanwhile, when only two of the asset classes are used, the withdrawal rate is 3.34 percent with an optimal asset allocation of 47.2 percent stocks and 52.8 percent bonds. The asset allocations provided in Exhibit 5.28 cannot be seen in the visual found in Exhibit 5.27, but they do exist behind the scenes within the software.

Exhibit 5.28

Sustainable Withdrawal Rates with 10% Failure for 30-Year Retirement Horizon

Asset Classes	Withdrawal Rate	Real Portfolio Return	Portfolio Standard Deviation	Asset Allocation
US Stocks	3.34%	3.4%	10.5%	47.2%
US Bonds				52.8%

With Five Asset Classes

Asset Classes	Withdrawal Rate	Real Portfolio Return	Portfolio Standard Deviation	Asset Allocation
US Stocks	3.61%	4.2%	11.1%	14.9%
US REITS				30.5%
MSCI EAFE				20.4%
US Bonds				34.2%
US Bills				0.0%

This case study illustrates how the framework for incorporating your own capital market expectations works. Of course, formulating appropriate capital market expectations is hard. Volatility especially complicates the process. A common view is that future stock returns will be lower than historical averages, but what about stock volatility? Would lower stock returns be accompanied by lower volatility, or is it reasonable to keep volatility the same? Might we even expect volatility to increase? The withdrawal rate lines in Exhibit 5.27 are upward sloping, making pinpointing the precise combination of expected